

Spring Boot Learning Guide

This guide is designed for beginners and Java developers who want to learn Spring Boot for backend and full stack web development.

1. Introduction to Spring Boot

- What is Spring Boot?
- Advantages of Spring Boot
- Spring vs Spring Boot
- Project Structure

2. Environment Setup

- Installing JDK
- Installing IntelliJ IDEA / VS Code
- Using Spring Initializr
- Maven and Gradle Basics

3. Core Spring Boot Concepts

- Dependency Injection
- Beans and Annotations
- @SpringBootApplication
- @Component, @Service, @Repository

4. Building REST APIs

- @RestController
- @RequestMapping
- GET, POST, PUT, DELETE APIs
- Path Variables and Request Parameters

5. Database Integration

- Spring Data JPA
- Hibernate Basics
- Connecting MySQL

- CRUD Operations

6. Advanced Concepts

- Exception Handling
- Validation
- DTOs
- Pagination and Sorting
- JWT Authentication

7. Testing and Deployment

- Unit Testing
- Postman API Testing
- Packaging JAR Files
- Deploying on Render

8. Projects

- Student Management System
- E-Learning Website Backend
- Employee Management System
- Fix It Now Service Backend

Recommended Learning Timeline

Week	Focus Area
1	Spring Boot Basics + Setup
2	REST APIs
3	Database + JPA + Hibernate
4	Authentication + Validation
5	Projects + Deployment

Recommended Tools:

- IntelliJ IDEA / VS Code
- MySQL Workbench
- Postman
- Maven
- Git & GitHub

Practice by building real-world backend projects and integrating frontend applications. Spring Boot is one of the most demanded backend technologies for Java developers.